Laboratoire 3 : Pont en H – Groupe B18

1. Alimenter le moteur en redressement synchrone

To work with “redressement synchrone” we will want to manipulate all transistors (T1, T2, T3, T4) given in the configuration below.

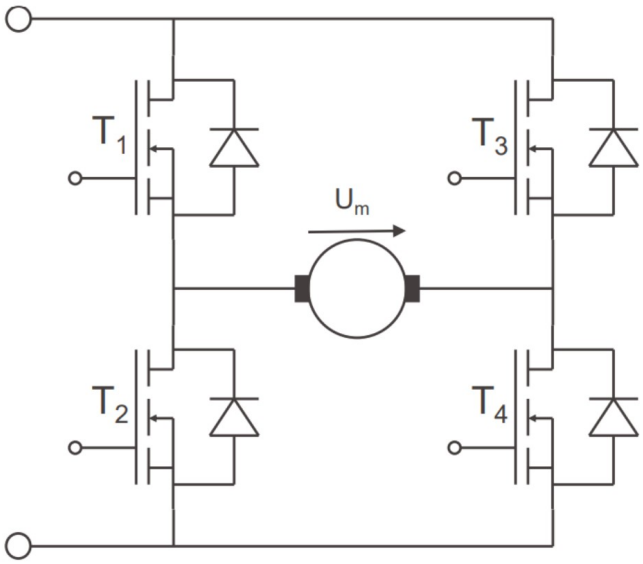


Figure 1: Schema for H-bridge (from class notes)

We observe the initial default state, where we have two PWM signals, both complementary of each other (separated by a dead-time, which is necessary to prevent a short-circuiting scenario wherein all transistors are on, and the current finds a path with less resistance that doesn’t go through the motor). One PWM signal controls the top transistors and the other the bottom transistors. When we have the same duty cycle on both of them we will switch both branches at the same time, which will result in a situation where the motor has net 0V differential across its terminals, resulting in no motion.

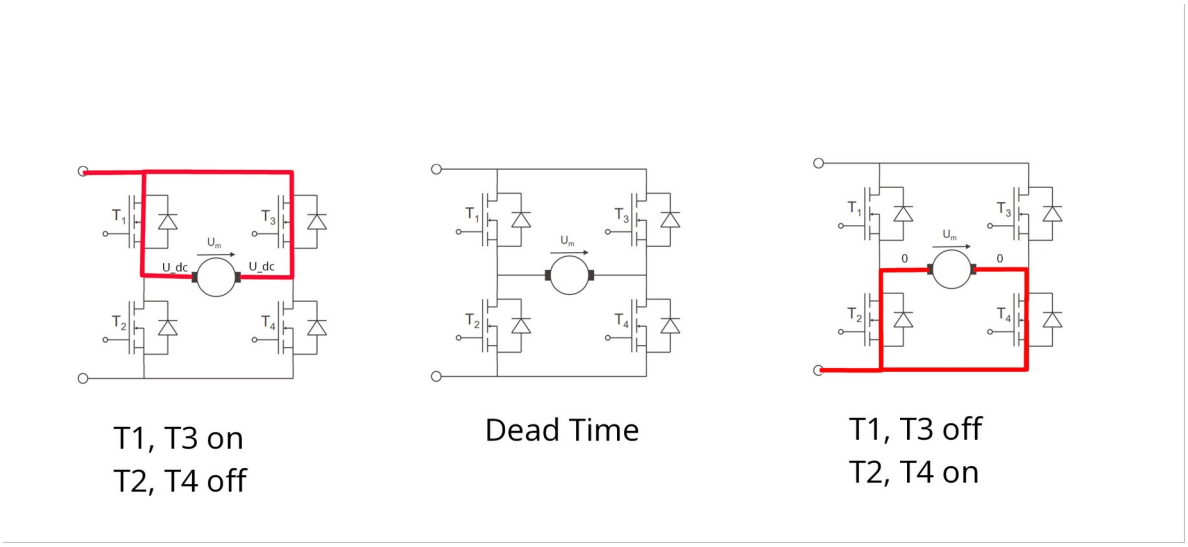


Figure 2: Demonstration when both branches have equal "taux"

We must ensure there is a potential difference across the motor for it to move, and we do this by varying the duty cycle of each branch individually so they are asymmetric and result in creating this imbalance. We do this as follows,

$$taux = (P/4096) * PWM_{period}$$

Knowing that (P) is the potentiometer reading we obtain, scaling it to its normalization range, as it ranges from 0 → 4096. We wish to ensure that at one extreme, when $P = 0$, we have all the voltage differential in one direction, and at $P = 4096$ in the opposite sense. We can obtain this relation by imposing the following duty cycles on the two branches, by altering their Compare Register, which dictates when the transistor will be commanded to turn on.

$$CCR1 = taux$$
$$CCR2 = PWM_{period} - taux$$

We get the following measurement on the oscilloscope, for both low and high loads, blue is voltage across motor and green is current:

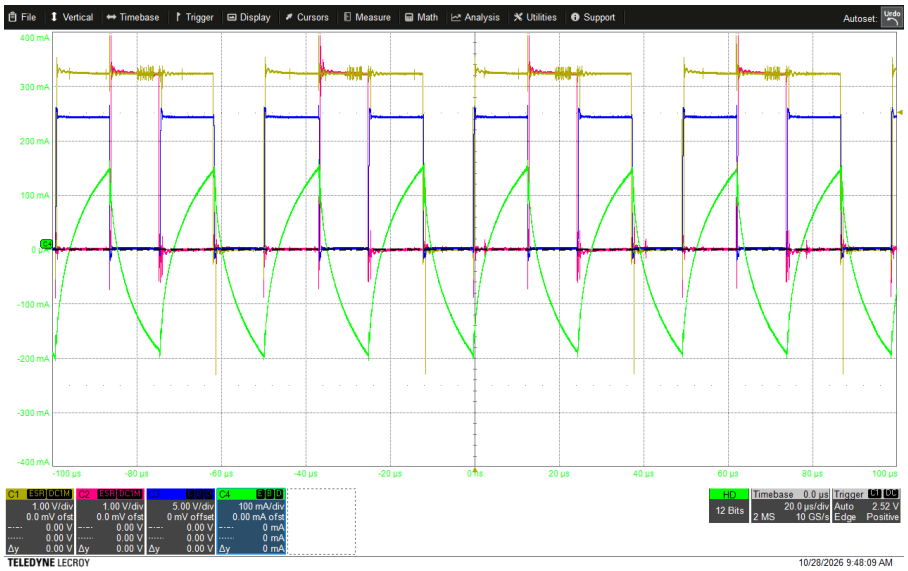


Figure 3: 25/75 split, with no external load

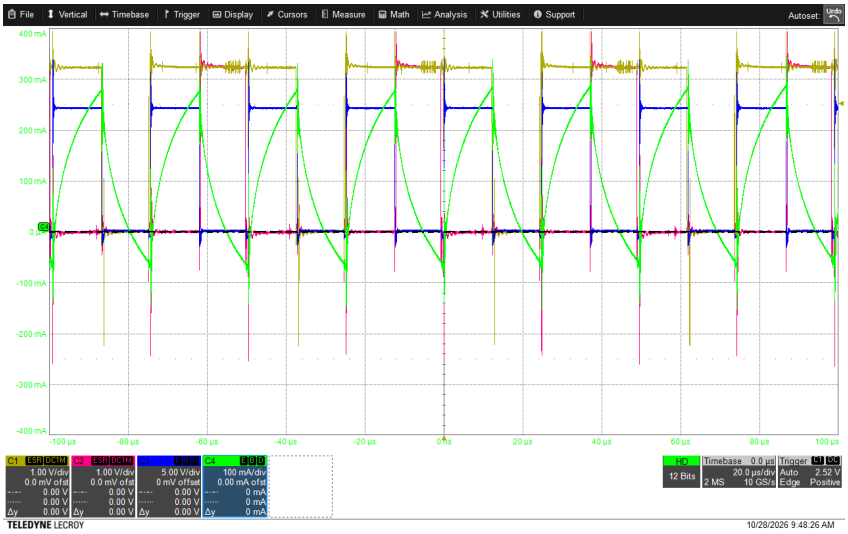


Figure 4: 25/75 split, with external load

A brief analysis of what is happening approximately above is shown below, Note that the colours are meant to represent which transistor is on, under the respective branch's PWM signal:

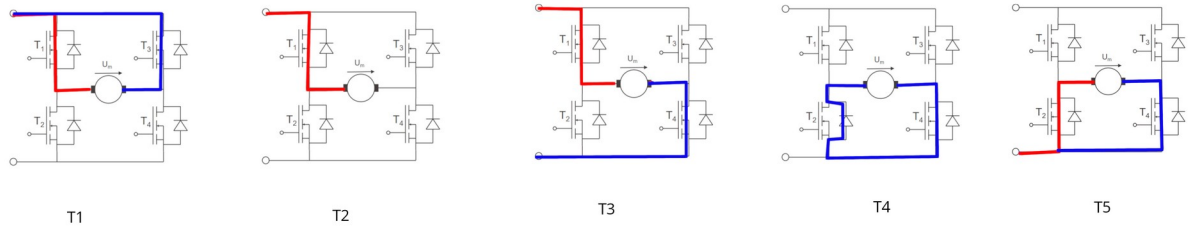


Figure 5: Diagram demonstrating when there is a asymetric difference bewteen Left and Right Branch Duty Cycle

- **Step 1:** Initially both top transistors are open, voltage difference across motor is 0.
- **Step 2:** Transistor 3 (top-right) closes as it has a smaller duty cycle, 25%. Dead time as Transistor 4 hasn't turned on yet
- **Step 3:** The transistor 4 (bottom right) turns on, and stays on. Now the motor has a voltage differential of + U_{DC}
- **Step 4:** The transistor 1 turns off, and momentarily the current has to flow through the freewheeling diode on T2 during the dead-time within the left branch.
- **Step 5:** The transistor 2 turns on, and the current can flow smoothly through the low-resistance transistor instead of the diode.

By controlling the duty cycle of the branches asymmetrically, we were able to replicate and ensure that there exists a on average potential difference across the motor, and that this will then result in the motor being able to operate. We can also observe the benefit with "redressement synchrone" is that for low load on the motor, the current will keep flowing, as it passes through the transistor and not the diode, gently going under 0A and then remounting, whereas with a diode it would shut off completely, meaning the motor speed feels smooth and not jerky at low loads.

2. Alimenter le moteur en PWM double

In "*PWM Double*" we have control over both diagonal transistors, either T1,T4 or T2,T3. In our case, we have T1 and T4 being manipulated. This is done by applying a mask to the Capture/Compare Register. This register is responsible for gating which transistors actually get the PWM signal that is being computed and created. Instead of manipulating the "*taux*" we manipulate these bits. In the "*PWM Double*" situation, we turn both diagonal transistors off / on, and when they are off, they have no free-path except through the freewheeling opposite diagonal diodes, to maintain the current direction flow of the motor, back into main power supply. As the current now flows from Ground (0V) to power supply (U_{DC}), creating a swing from + U_{DC} to -U_{DC}. This results in harsher spikes of current, as has to alternative within a larger range in the same time.

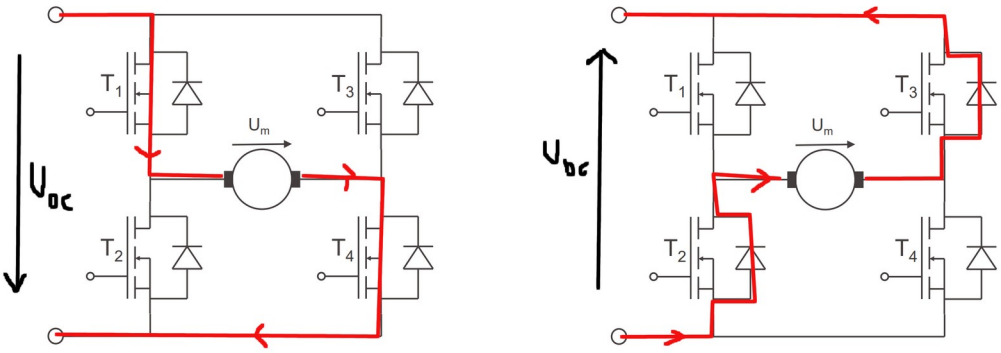


Figure 6: PWM Double turned on and off

In the figure above we see the cycle, from when the transistors are on, to when they are turned off (the motor still demands the current, and the inductance requires it, and the only available path is the one shown in the flow to the right.

An interesting thing we observed was when we put the potentiometer all the way to the left (i.e. longer duty cycle) the motor would stutter temporarily and then go back to rotating. The reason for this is the bootstrap did not have enough time to recharge. This was due to the rapid voltage spikes before settling, meaning the freewheeling diodes conduct but for a very small duration. The capacitor charges during this period, and as the time it has to recharge is so small, it doesn't have enough charge to last through higher duty cycles. Turning to the other side doesn't affect because you are keeping a low duty cycle, hence no problem.

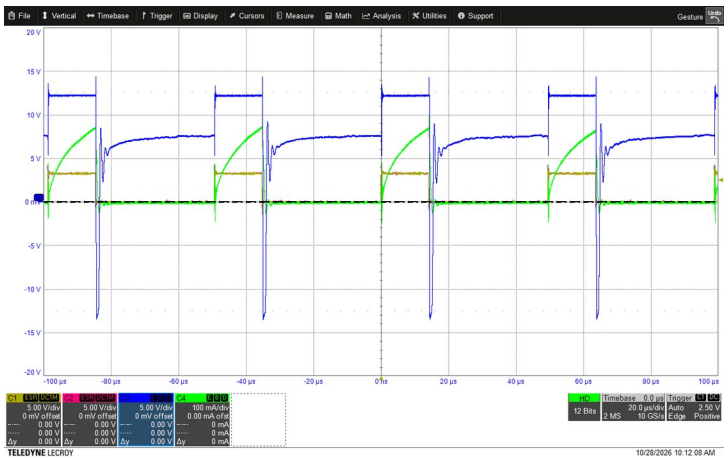
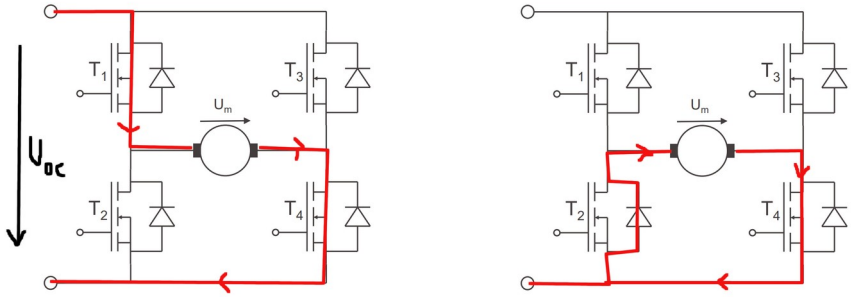


Figure 7: PWM Double with Potentiometer on 50%

3. PWM Simple

In "*PWM Simple*" we have control over one transistor, while the other is always on. Interestingly to note, given our bootstrap observation previously, we do not have the same problem here. The reasoning can be explained by the fact that Voltage drops to 0 not -12V, as current instead of flowing through the opposite diagonals diodes, flows through the loop at the bottom, as shown bellow.



This means voltage drops to 0V and not -12V during the same cycle, resulting in a smoother, longer climb, less violent current spikes, and consequently less energy loss. This extra time gives the capacitor longer to charge up, and hence it can work at higher duty cycles, which we observed by putting a high duty cycle and motor did not stutter.

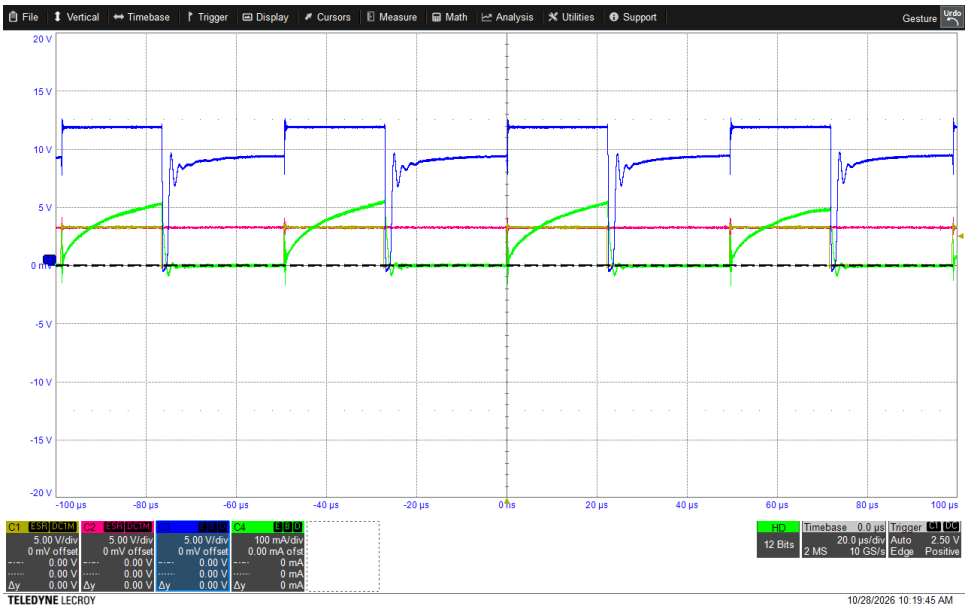


Figure 8: PWM-Simple, with potentiometer at the center

4. Tourner moteur dans le deux sens (en PWM Simple)

Inherently in the PWM Simple case, we only alternate between Source Voltage and 0V, meaning the motor only ever turns in one direction. We want to make it turn in both. The idea here is we need to use all transistors, and adapt the potentiometer so when it is at 50%, this is the stopping point for the motor. We need to adapt our scaling and usage of the potentiometer value, to switch between the branches that are active and also their duty cycle, so that according to where the potentiometer is, we will rotate the motor in one direction and in another direction.

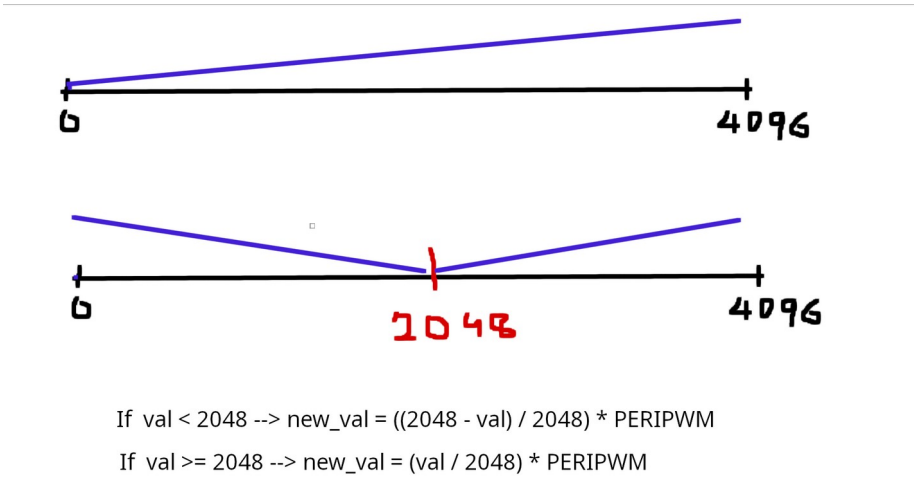


Figure 9: Demonstration on how to convert Potentiometer values to turn motor both sides in PWM Simple

Now some observations with the oscilloscope,

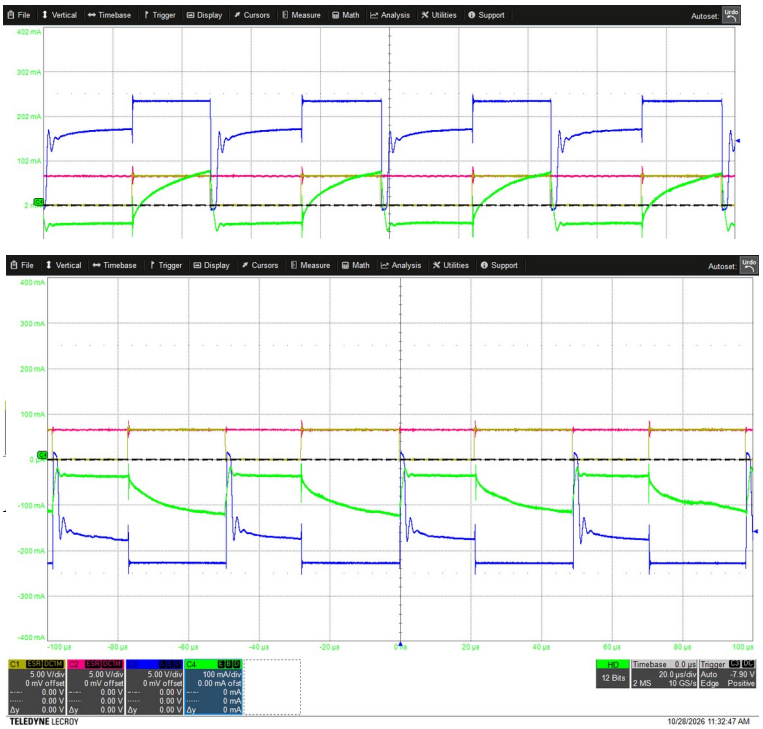


Figure 11: Potentiometer at 75%, T2 and T3 active

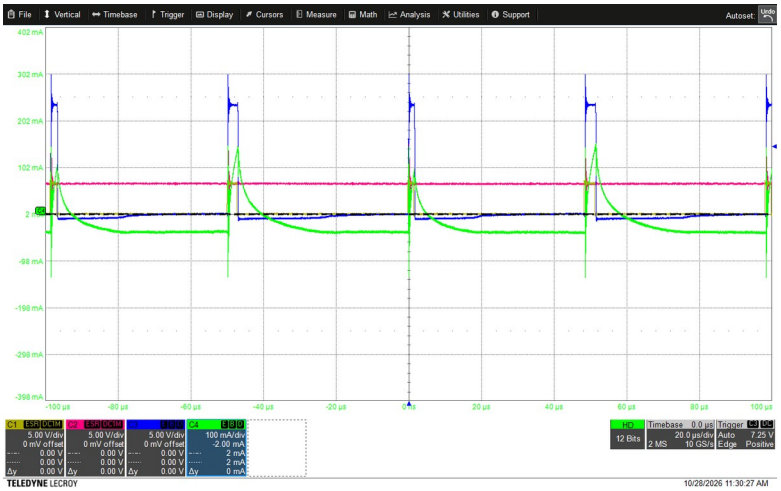


Figure 12: When Potentiometer is at 50% viewed by T1 and T4

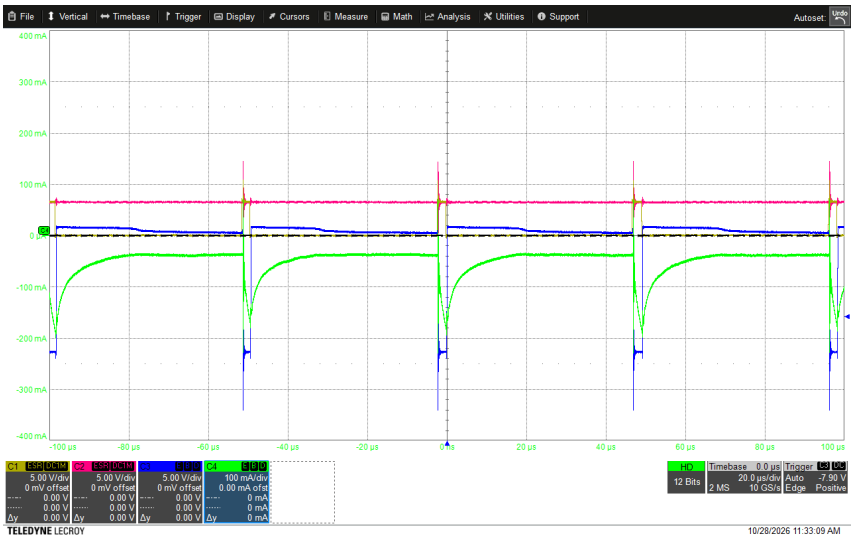


Figure 13: When Potentiometer is at 50% viewed by T2 and T3